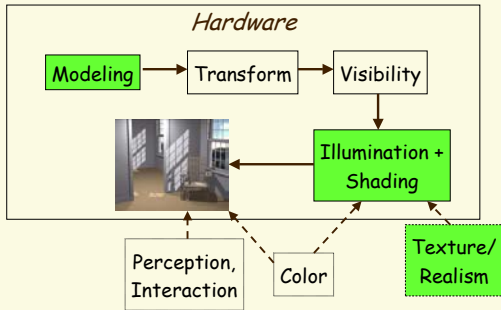


Rendering Pipeline



Reading

- Chapter 6 & 7 of "3D Computer Graphics" by Watt
- Chapter 6 & 13 of "Interactive Computer Graphics" by Angel
- Chapter 16 of "CG: Principles & Practice" by Foley, van Dam et al.
- Chapters 11, 12, 13, 15 "Principles of Digital Image Synthesis" by Glassner

Illumination / Shading

- Distinction between illumination and shading models
 - **illumination** - calculate intensity at a point on surface
 - **shading** - uses calculated intensities to shade polygons (uses illumination models)
- we'll review the important models

Illumination / Shading

- We'll talk more about concepts that lead to realism:
 - global illumination:
 - ray tracing + radiosity
 - "special effects/tricks":
 - shadows, texture maps, bump maps, anti-aliasing, transparency, reflection maps, refraction

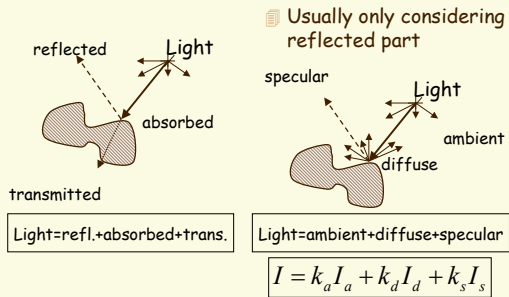
Local Illumination

- Local vs. global illumination models
 - local (typically) - how is one point of the scene illuminated directly by the light source
 - is light source only source of illumination?
 - Simple models lump the rest into a single ambient term
 - do not account for reflections within the environment

Local Illumination

- Local vs. global illumination models
 - global - illuminates the whole scene
 - typically makes use of local illumination model
 - incorporates inter-reflectance of objects

Light Effects



Diffuse Light

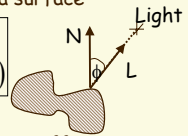
- Light absorbed by the surface and then reflected equally to all directions
- Models dullness, roughness of a surface

Lambert's Law:
(perfectly diffuse surface)

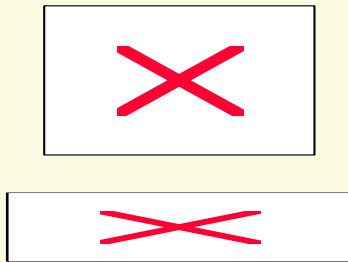
$$I = k_d I_d \cos \phi$$

$$= k_d I_d (N \cdot L)$$

- I_d : intensity of light source
- k_d : material's diffuse reflection coefficient
- N : normal vector (normalized)
- L : light source vector (normalized)



Diffuse Light



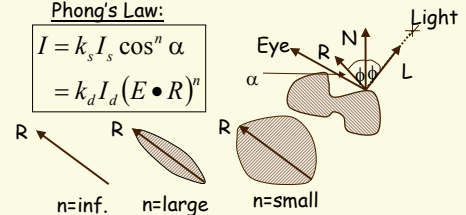
Specular Light

- Light that is reflected from the surface unequally to all directions
- Models reflections on shiny surfaces

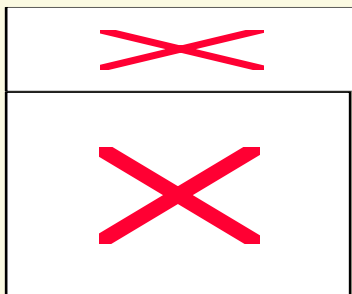
Phong's Law:

$$I = k_s I_s \cos^n \alpha$$

$$= k_s I_d (E \cdot R)^n$$



Specular Light



Ambient Light

- is the light in the environment evenly reaching all surfaces from all directions
- light location doesn't matter
- eye position doesn't matter

$$I = k_a I_a$$

- I_a : ambient light
- k_a : material's ambient reflection coefficient

Ambient Light



- I_A : ambient light

$$I = k_a I_a$$
- k_a : material's ambient reflection coefficient
- Models general level of brightness in the scene
- Accounts for light effects that are difficult to compute (secondary diffuse reflections, etc)

Depth Cueing/Illumination

- Theory - illumination fall-off with square of distance $1/d^2$
- doesn't give good results
- often a factor of $1/d$ or $1/(d+c)$ is used
- a way to simulate fog - linear fade ...
- use start s and end e fade distances
- d = distance from viewer:

$$\begin{aligned}
 I &= I_0 && \text{for } d < s \\
 &= \frac{(e-d)}{(e-s)} I_0 + \frac{(d-s)}{(e-s)} I_f && \text{for } s < d < e \\
 &= I_f && \text{for } d > e
 \end{aligned}$$

Depth Cueing/Illumination

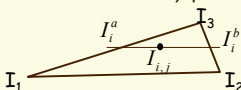


Shading

- Illumination Model:** determine the color of a surface (data) point by simulating some light attributes.
- Local IM:** deals only with isolated surface (data) point and direct light sources.
- Global IM:** takes into account the relationships between all surfaces (points) in the environment.
- Shading Model:** applies the illumination models at a set of points and colors the whole scene.

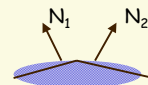
Shading - efficiently

- Constant Shading:**
- compute illumination at one point of the primitive (e.g. surface) and apply it for the whole primitive.
- Interpolated Shading:**
- compute illumination at borders (e.g. vertices) of the primitive and interpolate the color
- Accurate Shading:**
- compute illumination at *every* point of the primitive



Flat Shading

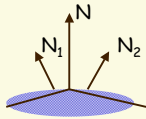
- Polygon meshes approximate smooth curved surfaces with planar facets. Using the previous methods does not generate an illusion of smooth curved surface.



- Reason:** discontinuity of the normal vectors.

Gouraud Shading

- Assign vertex the normal of the smooth surface.
Or
- Average the normal of all neighboring polygons



- Interpolate colors along edges and scan-lines

Gouraud Shading

Flat Shading

Gouraud Shading



Phong Shading

- Gouraud Shading does not properly handle specular highlights.



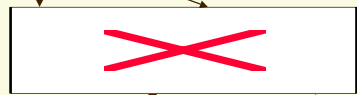
- Reason:** Colors are interpolated

Solution:

- Compute averaged normal at vertices (Gouraud)
- Interpolate normals along edges and scan lines!
- Apply illumination model at every pixel

Phong Shading

Gouraud Shading



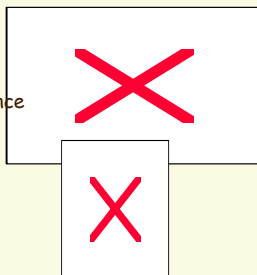
Phong Shading

Shading - problems

Discontinuities!!

E.g.

- polygonal silhouette
- orientation dependence
- shared vertices

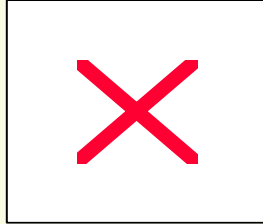


Transmission

- Objects can transmit light!
- Light may be transmitted specularly (transparent) or diffusely (translucent), like reflected light
 - specular transmission is exhibited by materials such as glass
 - diffuse transmission is exhibited by material which scatter the transmitted light (translucent, e.g. frosted glass, church)

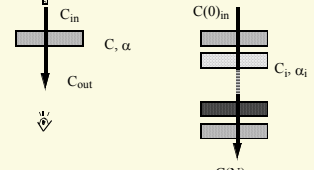
Transparency

- 2 types of transparency - nonrefractive and refractive (more realistic)



Non-Refractive Transparency

- Requires sorting! (BTF vs. FTB)
- "over" operator - Porter & Duff 1984



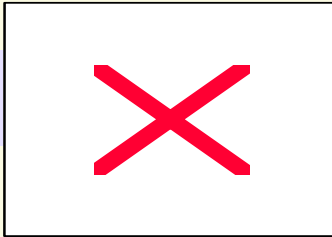
$$C_{out} = C_{in} \cdot (1 - \alpha) + C \cdot \alpha$$

$$C(i)_{in} = C(i-1)_{out}$$

Refractive Transparency

- Transmitted light may be refracted according to Snell's law!

$$\frac{\sin q_i}{\sin q_t} = \frac{\eta_t}{\eta_i}$$



Refractive Transparency

- η_t, η_i - indices of refraction (ratio of speed of light in vacuum to the speed of light in the medium)
- depend on:
 - temperature
 - wavelength of light
 - a good book :)
- hence - refracted light can have a different color

$$\frac{\sin q_i}{\sin q_t} = \frac{\eta_t}{\eta_i}$$

Refractive Transparency

- In general - refraction effects are ignored in simple renderers (e.g. scanline z-buffer)
- most materials in scene are non-transmitting (so it's ok most of the time)
- Extension to simple model:

$$I = k_a I_a + k_d I_d + k_s I_s + k_t I_t$$
- k_t - transmission coefficient of a material
- I_t - intensity of transmitted light

Implementation - an example

- Modify the z-buffer so that it maintains a depth-sorted list of visible surfaces at each pixel (A-buffer)
- for each polygon:
 - if its in front of nearest opaque surface
 - if its opaque
 - it becomes the nearest opaque surface (I.e. insert in the list and delete everything behind it)
 - else
 - insert by z into the list
 - else ignore it

Implementation - an example

- Blend two polygons like in the non-refractive version

$$C_{out} = C_{in} \cdot (1 - \alpha) + C \cdot \alpha$$

Implementation - an example

- linear approximation doesn't model curved surfaces well. It does not account for the increased thickness of the material at silhouette edges, which reduces transparency.
- To approximate these effects - use a non-linear transparency factor:

$$\alpha = \alpha_{\max} + (\alpha_{\min} - \alpha_{\max}) \cdot \left[1 - (1 - |N_z|)^p \right]$$

Implementation - an example

$$\alpha = \alpha_{\max} + (\alpha_{\min} - \alpha_{\max}) \cdot \left[1 - (1 - |N_z|)^p \right]$$

- $\alpha_{\min}, \alpha_{\max}$ - minimum and maximum transparencies for the surface
- N_z - is the z component of the surface normal
- p - is a modelling parameter (transparency power factor)

Total internal reflection

- When light passes from one medium into another whose index of refraction is lower the transmitted angle is larger than the incident angle
- at one point the light gets reflected - called total internal reflection

Physically Based Illumination

- Everything so far has been pretty heuristic
- cannot model:
 - wavelength dependent phenomena
 - anisotropic behaviours
 - many other physical phenomena (real physics)
- ongoing research - main contributions
 - Blinn (1977), Cook, Torrance (1982)
 - Kajiya (1985), Cabral et al (1987)
 - Hanrahan, Krüger (1993)

Physically Based Illumination

- flux: the rate at which light energy is emitted, measured in watts (W)
- solid angle: angle at the apex of a cone
- steradian (sr): the solid angle such that the cone intercepts an area equal to the square of the sphere's radius. There are 2 pi steradians in a hemisphere.
- radiant intensity: flux radiated into a unit solid angle (W/sr). Intensity, referring to a point source.

Physically Based Illumination

- Radiance I : radiant intensity per unit foreshortened (or projected) surface area ($W / (sr \cdot m^2)$) Intensity, referring to a surface.
- irradiance (flux density) E : incident flux per (unforeshortened) unit surface area (W / m^2)

$$E_i = I_i (N \cdot L) d\omega_i \\ = I_i \cos \theta_i d\omega_i$$

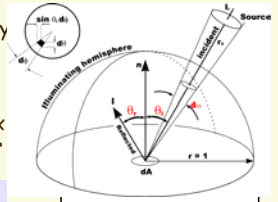


Physically Based Illumination

- bidirectional reflectivity (BRDF) ρ : ratio of reflected radiance (intensity) to the incident irradiance (flux density) responsible for it

$$\rho = \frac{I_r}{E_i}$$

$$I_r = \rho I_i (N \cdot L) d\omega_i$$



Physically Based Illumination

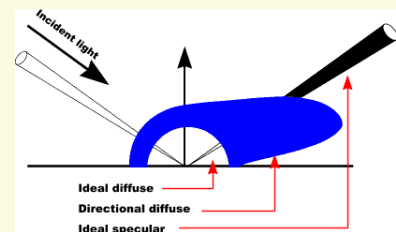
- Typical model: BRDF is part diffuse and part specular reflection!

$$\rho = k_d \rho_d + k_s \rho_s \\ k_d + k_s = 1$$

- Take into account light from other SURFACES! + an ambient component:

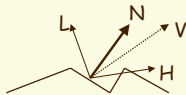
$$I_r = \rho_a I_a + \sum_{1 \leq j \leq n} I_{i_j} (N \cdot L_j) (k_d \rho_d + k_s \rho_s) d\omega_i$$

Physically Based Illumination



Torrance-Sparrow

$$\rho = \frac{F_\lambda}{\pi} \frac{DG}{(N \cdot V)(N \cdot L)}$$



- D** - Microfacet Distribution Function
 - how many "cracks" do we have that point in our (viewing) direction?
- G** - Geometrical Attenuation Factor
 - light gets obscured by other "bumps"
- F** - Fresnel Term
 - which portion of the incoming light gets reflected?

TS - Microfacet Distribution Function

- Microfacet Distribution Function D

- how straight is our surface? - or -
- measure of scattering due to variation in angle of microfacets
- a statistic approximation, I.e. need a statistic distribution function
- many possibilities:
- Blinn
- Beckmann (most

$$D = e^{-\left(\frac{\alpha}{m}\right)^2}$$

$$D = \frac{1}{4m^2 \cos^2 \alpha} e^{-\left(\frac{\tan \alpha}{m}\right)^2}$$

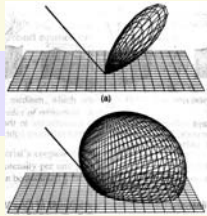
TS - Microfacet Distribution Function

many possibilities:

- Blinn $D = e^{-\left(\frac{\alpha}{m}\right)^2}$

- Beckmann (most effective)

$$D = \frac{1}{4m^2 \cos^2 \alpha} e^{-\left(\frac{\tan \alpha}{m}\right)^2}$$



TS - Geometrical Attenuation Factor

Geometrical Attenuation Factor G

- how much are the "cracks" obstructing themselves?



No interference

shadowing

masking

$$G = \min \left[1, \frac{2(N \cdot H)(N \cdot V)}{(V \cdot H)}, \frac{2(N \cdot H)(N \cdot L)}{(V \cdot H)} \right]$$

TS - Fresnel Term

Fresnel Term F

- basically - some light gets refracted and some gets reflected - which is the reflected part???
- "... specifies the ratio of reflected light from a dielectric (nonconducting) surface as:"

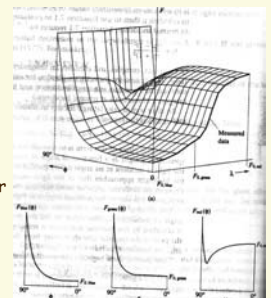
$$F_{\lambda} = \frac{1}{2} \left(\frac{\tan^2(\theta_i - \theta_t)}{\tan^2(\theta_i + \theta_t)} + \frac{\sin^2(\theta_i - \theta_t)}{\sin^2(\theta_i + \theta_t)} \right)$$

- perfectly opaque $\Rightarrow \theta_t = 0$; and hence $F = 1$

TS - Fresnel Term

Example - Copper

- color shift as θ goes from 0 to $\pi/2$
- at grazing, specular highlight is color of light
- typically its always the color of light, no matter what angle



TS - Summary

Example - Copper

- color shift as θ goes from 0 to $\pi/2$
- at grazing, specular highlight is color of light
- typically its always the color of light, no matter what angle

